QP CODE : 24900066

Mahatma Gandhi University, Kottayam

First Semester MGU - UGP(Honours) regular Examination November 2024

Core Course - MG1CCRBCA101 - DISCRETE MATHEMATICS

(2024 ONWARDS)

SCHEME OF VALUATION

PART A

Answer any 5 Questions, Each Question Carries 2 marks.

1. The Conjunction of p ∧ q is the proposition p and q (similar statement) - 1 mark
   Truth table - 1 mark (1 + 1 = 2 marks)
2. Heather will not go to the concert and Steve will not go to the concert - 2 marks
3. a) { 1,2,3……..9 } - 1 mark ,
   b) { x / x is a positive integer less than 10 } - 1 mark, (1 + 1 = 2 marks)
4. 1) {1,2,3,4,7,8,9,10} - 1 mark , 2) {2,4,5,6,7} - 1 mark, (1 + 1 = 2 marks)
5. For writing **any of** not reflexive or not transitive - 1 mark
   ∴ not an equivalence relation - 1 mark (1 + 1 = 2 marks)
6. Definition of directed graph - 1 mark, example - 1 mark (1 + 1 = 2 marks)
7. For finding | A | = 0 - 2 marks
8. A^{-1} =

$$\frac{1}{18}\begin{bmatrix} 5 & 1 \\ 2 & 4 \end{bmatrix}$$

   1 mark , solution x = ⅓ y = ⅓ 1 mark (1 + 1 = 2 marks)

PART B

Answer any 5 Questions, Each Question Carries 6 marks.

9. a) proposition, 8 is not a prime number or it is not the case that 8 is a prime number
   b) proposition c) proposition, 9 not greater than 4 or similar - 2 marks each
   (2 x 3 = 6 marks)
10. a) Modus ponens - 2 marks, Explanation - 1 mark (3 marks)
    b) Simplification - 2 marks, Explanation - 1 mark (3 marks)
11. For Finding A x B - 2 ½ marks , For Finding B x A - 2 ½ marks
    A x B ≠ B x A - 1 mark (2 ½ + 2 ½ + 1 = 6 marks)
12. a) Number of bits in the string - 1 ½ marks , length is 11 - 1 ½ marks
    b) 11111110 3 marks
13. $2^{n(n-1)}$. Stage wise marks can be fixed by the examiner - 6 marks
14. Not symmetric since the matrix is is not symmetric or 1st row and 1st column are not the same (OR Not transitive by showing any transitive pairs like (2,3) ∈ R, (3,1) ∈ R but (2,1) ∉ R) - 3 marks for any of these explanation
    ∴ not an equivalence relation - 3 marks (3 + 3 = 6 marks)
15. x= -3 , y = $\frac{-13}{2}$, z = 4, a = 3 - 1 ½ marks for each correct answer (1 ½ x 4=6 marks)
16. |A| = - 2, $|A_1|$ = - 2, $|A_2|$ = - 4, $|A_3|$ = - 6 , 1 Mark each(1 x 4 = 4 marks)
    Formula - ½ mark, x = 1, y = 2, z = 3 - ½ marks for each correct answer
    (½ x 4=2 marks) (4 + 2 = 6 marks)

PART C

Answer any 3 Questions, Each Question Carries 10 marks.

17. a) Conditional and bi conditional statements .Definition and example 2 ½ each
(2 x 2 ½ = 5 marks)

b) Contrapositive - If the home team doesn't win, it's not raining.
Converse - If the home team wins, then it's raining
Inverse - If it's not raining, the home team doesn't win (2 + 1 ½ + 1 ½ = 5 marks)

18. a) Definition of Tautology - 3 marks

b)

p	q	$\neg p$	$p \rightarrow q$	$\neg p \wedge (p \rightarrow q)$	$\neg q$	$(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$
T	T	F	T	F	F	T
T	F	F	F	F	T	T
F	T	T	T	T	F	F
F	F	T	T	T	T	T

.Qn is wrong .It is not a Tautology. For a reasonable attempt give 5 marks. For correct table give 7 marks (3 + 7 = 10 marks)

19. Distributive laws

1. A ∧ (B∨C) = (A ∧ B) ∨ (A ∧ C)

A	B	C	(B∨C)	A ∧ (B∨C)	(A ∧ B)	(A ∧ C)	(A ∧ B) ∨ (A ∧ C)
T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	T
T	F	T	T	T	F	T	T
F	T	T	T	F	F	F	F
T	F	F	F	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

2. A ∨ (B ∧ C) = (A ∨ B) ∧ (A ∨ C)

A	B	C	(B∧C)	A ∨ (B∧C)	(A ∨ B)	(A ∨ C)	(A ∨ B) ∧ (A ∨ C)
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T

F	T	T	T	T	T	T	T
T	F	F	F	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Associative laws : 1. A ∧ (B ∧C) = (A ∧ B) ∧ C

A	B	C	(B ∧C)	A ∧ (B∧C)	(A ∧ B)	(A ∧ B) ∧ C
T	T	T	T	T	T	T
T	T	F	F	F	T	F
T	F	T	F	F	F	F
F	T	T	T	F	F	F
T	F	F	F	F	F	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

2. A ∨ (B ∨ C) = (A ∨ B) ∨ C

A	B	C	(B ∨ C)	A ∨ (BVC)	(A ∨ B)	(A ∨ B) ∨ C
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	T	T
F	T	T	T	T	T	T
T	F	F	F	T	T	T
F	T	F	T	T	T	T
F	F	T	T	T	F	T
F	F	F	F	F	F	F

2 ½ Marks each (2 ½ x 4 = 10 Marks)

20.

$$M_{R_1} \cup M_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \qquad M_{R_1} \cap M_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$M_{R_1 \circ R_2} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \qquad M_{R_2 \circ R_1} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

2 ½ Marks each (2 ½ x 4 = 10 Marks)

21.

$$(A+B)^2 = \begin{bmatrix} 18 & 4 & 5 \\ 28 & -8 & 20 \\ 28 & -4 & 6 \end{bmatrix} \quad A^2 = \begin{bmatrix} 5 & 1 & 3 \\ 2 & 7 & 4 \\ 2 & 2 & 7 \end{bmatrix} \quad B^2 = \begin{bmatrix} 13 & -6 & 3 \\ 8 & -6 & 4 \\ 20 & -10 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} -1 & 2 & 3 \\ 18 & -11 & 8 \\ 8 & -6 & 6 \end{bmatrix} \quad BA = \begin{bmatrix} 1 & 7 & -4 \\ 0 & 2 & 4 \\ -2 & 10 & -9 \end{bmatrix}$$

For any correct 3 matrices give 3 x 2 = 6 marks, for the rest 2 matrices give 2 x 1 ½ = 3 marks

$$A^2 + B^2 + AB + BA = \begin{bmatrix} 18 & 4 & 5 \\ 28 & -8 & 20 \\ 28 & -4 & 6 \end{bmatrix}$$ = $(A+B)^2$, give 1 mark, (6+3+1=10 marks)

22. $|A| = 1$ 1 mark

$$A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix} \quad A^2 = \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

Give 3 marks for each matrix. 3 x 3 = 9 marks (1 + 9 = 10 marks)

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